Human Review: Borel Factor-Subspace Selection

Original automatic proof: `solution_packet.pdf`
Checked date: 19 June 2026
Status: Partially manually checked; author verification recommended.
Verdict: The proof looks right, but I am not able to verify every technical detail to the appropriate level.

Human Comments

The proposed argument addresses Question 6.1 of Antunes–Beanland–Braga, *Uniformly factoring weakly compact operators and parametrised dualisation.* The question asks whether the Borel measurable versions of Theorem 5.1, Corollary 5.12, and Corollary 5.13 hold.

The proof appears to make one precise repair to the published proof of Theorem 5.1. In the original argument, after constructing the Borel code map

$$\eta = \psi \circ \sigma : \mathcal{B} \longrightarrow \mathbb{N}^{\mathbb{N}},$$

the authors choose a closed tree whose projection is the analytic set $\eta(\mathcal{B})$. A branch above each code is then chosen using Jankov–von Neumann uniformization, which gives only $\sigma(\Sigma^1_1)$-measurability. The proposed proof observes that $\eta(\mathcal{B})$ is not an arbitrary analytic set: it is the image of a Borel map. It therefore builds the closed tree from a closed parametrization of the Borel graph of η, obtaining a genuinely Borel branch selector.

The central descriptive-set-theoretic lemma looks correct. Given a Borel map $\eta : S \to \mathbb{N}^{\mathbb{N}}$, one first codes the standard Borel space S as a Borel subset of Baire space, considers the Borel graph

$$G = \{(i(s), \eta(s)) : s \in S\},$$

and represents this Borel set as a one-to-one continuous image of a closed set. This gives a closed set

$$C \subseteq \mathbb{N}^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$$

and a Borel lift $\tau : S \to C$ with $p(\tau(s)) = \eta(s)$. This seems to supply exactly the missing Borel branch choice.

The key structural check also appears to work. Every branch of the new tree still has first coordinate equal to $\eta(A_0) = \psi(\sigma(A_0))$ for some original parameter $A_0 \in \mathcal{B}$. Thus the modified tree should not introduce any new "bad" branch spaces. The reflexivity argument in the Kurka amalgamation step is therefore the same as in the source proof: the branch spaces are reflexive, and Kurka's proposition then gives the reflexivity of the amalgamated space Z.

The measurability repair also appears sound. In the source proof, the proof of Claim 5.11 explicitly reduces the measurability of Φ to the measurability of the assignment $A \mapsto t_A$, together with already Borel objects such as $\sigma(A)$, φ_A, and the fixed selectors. Replacing the old $\sigma(\Sigma^1_1)$-measurable selector by the Borel selector $t_A = \tau(A)$ should therefore upgrade the same displayed formula from $\sigma(\Sigma^1_1)$-measurability to Borel measurability. The Borelness of $\Psi(A) = \overline{\operatorname{span}}\{\Phi(A, d_n(X)) : n \in \mathbb{N}\}$ then follows from the usual Kuratowski–Ryll-Nardzewski selectors and the standard Borel closed-span operation.

Citation Correction

There is a small citation-numbering correction in the automatic proof. The rational-code map

$$\psi : \mathrm{mbs} \to \mathbb{N}^{\mathbb{N}}$$

should be cited as coming from Lemma 5.9 of Antunes–Beanland–Braga, not Lemma 5.10. In the source paper, Lemma 5.9 is the result defining and recording the properties of ψ; Claim/Lemma 5.10 in that proof is the later reflexivity claim for the interpolation space Z.

Caveat

Although the method appears solid and the single proposed modification seems to target exactly the obstruction in the published proof, I am not sufficiently familiar with all the descriptive-set-theoretic and parametrized Banach-space machinery involved to certify the proof at the level one would want before making a firm mathematical claim. In particular, small technical issues could still be hidden in the passage from the modified closed tree to the exact Kurka amalgamation framework, or in the precise Borel coding of the resulting maps.

Review Outcome

This should be treated as a promising and apparently correct solution, but not as fully human-certified at this stage. I recommend sending the argument to the authors to ask whether the Borel graph/closed-tree replacement indeed removes the $\sigma(\Sigma^1_1)$-measurability obstacle in Question 6.1.